

Data Science Report: Market Sentiment & Trader Behavior

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1. Executive Summary

This project investigates the correlation between broad market sentiment (measured by the Fear & Greed Index) and individual trader performance on the Hyperliquid exchange. By analyzing a dataset of over **211,000 historical trades**, we aimed to identify whether following the crowd or trading against it yields better profitability.

Our analysis confirms a strong **Contrarian Signal**: Traders who opposed the prevailing market sentiment (e.g., Shorting during "Greed") significantly outperformed those who followed the trend.

2. Methodology

Data Preprocessing

- **Sources:** Hyperliquid Trader Data (Transactions) and Bitcoin Fear & Greed Index.
- **Merging:** Datasets were aligned on a daily timeframe. Timestamps were normalized to UTC dates to ensure accurate mapping of sentiment classifications to specific trades.
- **Cleaning:** "Size USD" was used as a proxy for risk/conviction, as direct leverage data was inconsistent or missing in the raw export.

Strategy Classification

Trades were categorized into three distinct behaviors based on the sentiment at the time of execution:

1. **Contrarian (Short on Greed):** Opening Short positions when sentiment is "Greed" or "Extreme Greed."
2. **Contrarian (Long on Fear):** Opening Long positions when sentiment is "Fear" or "Extreme Fear."
3. **Trend Following:** Longing during Greed or Shorting during Fear.

3. Key Findings

Insight A: "Fading the Crowd" Maximizes Profit

The most distinct signal discovered in the data is the superior performance of contrarian

strategies.

Strategy Type	Trade Count	Mean PnL (\$)	Median PnL (\$)
Contrarian (Short on Greed)	47,779	\$85.03	\$0.08
Contrarian (Long on Fear)	41,205	\$56.02	\$0.00
Trend Following	122,234	\$31.77	\$0.00

Observation:

- **Shorting on Greed** was the most profitable strategy, generating nearly **2.7x higher average returns** per trade compared to Trend Following.
- **Trend Following** was the most common behavior (comprising ~58% of all trades) but offered the lowest average edge.

Insight B: Conviction & Sentiment

(Refer to Figure 2 in Visual Evidence)

Analysis of Size USD indicates that traders tend to allocate capital differently based on sentiment. [Add a specific observation here looking at your 'size_usd_vs_sentiment.png', e.g., "Traders often take larger position sizes during Extreme Greed, potentially indicating overconfidence."]

4. Visual Evidence

Figure 1: Strategy Profitability

Figure 1 displays the distribution of Closed PnL across strategies. Note the higher upper-quartile performance for the Contrarian strategies.

Figure 2: Risk Appetite (Trade Size)

Figure 2 illustrates the average trade size in USD across different sentiment classifications. Since leverage data was unavailable, Size USD serves as the primary proxy for trader conviction.

5. Conclusion & Recommendations

The historical data suggests that retail trader profitability is negatively correlated with crowd alignment.

1. **Strategic Signal:** Automated strategies should weigh "Greed" sentiment as a strong

indicator to look for Short setups.

2. **Risk Management:** Caution is advised during "Trend Following" setups, as the average expected value (EV) is significantly lower than contrarian setups.